Integração por substituição

$F, \quad f: I \to \mathbb{R} \qquad F'(x) = f(x)$

$$\int f(x)\,dx = F(x) + C$$

$\varphi : J \to I$ bijetiva, derivável

$$\Phi(x) = F(\varphi(x))$$

$$\Phi'(x) = (F(\varphi(x)))' = F'(\varphi(x)) \cdot \varphi'(x)$$
$$= f(\varphi(x))\,\varphi'(x)$$

Exercício 6.3)

e - $\displaystyle\int \frac{e^{2x}}{3+e^x}\,dx$

$f : \mathbb{R} \to \mathbb{R}$, $x \mapsto \dfrac{e^{2x}}{3+e^x}$ $\qquad \varphi : \mathbb{R}^+ \to \mathbb{R}$, $x \mapsto \ln t$

$\varphi^{-1}(t) = e^x \qquad \varphi'(t) = \dfrac{1}{t}$

$$\int f(\varphi(t))\,\varphi'(t)\,dt = \int \frac{t^2}{3+t} \cdot \frac{1}{t}\,dt$$

$$= \int \frac{t}{3+t}\,dt = \int \frac{(t+3)-3}{3+t}\,dt = \int dt - 3\int \frac{1}{3+t}\,dt$$

$$= \underbrace{t - 3\ln|t+3|}_{\Phi(t)} + C$$

$$\Phi(\varphi^{-1}(x)) = e^x - 3\ln(e^x+3) + C$$

Maneira complicada de fazer do Fernando

Resolução mais rápida e fácil

$$\int \frac{e^{2x}}{3+e^x}\,dx = \int \frac{t^2}{3+t}\,\frac{1}{t}\,dt = \dots = t - 3\ln(t+3) + C$$

$$= e^x - 3\ln(e^x+3) + C$$

$e^x = t$

$\Leftrightarrow x = \ln t$

$\Leftrightarrow 1 \times dx = \frac{1}{t} \times dt$